

RSTE -- how one forward pass actually works

weight-tied over a nested-dissection separator tree; the SAME Merge / Bcast / Gate tensors act at every level and every node

why the parameter count does not depend on N

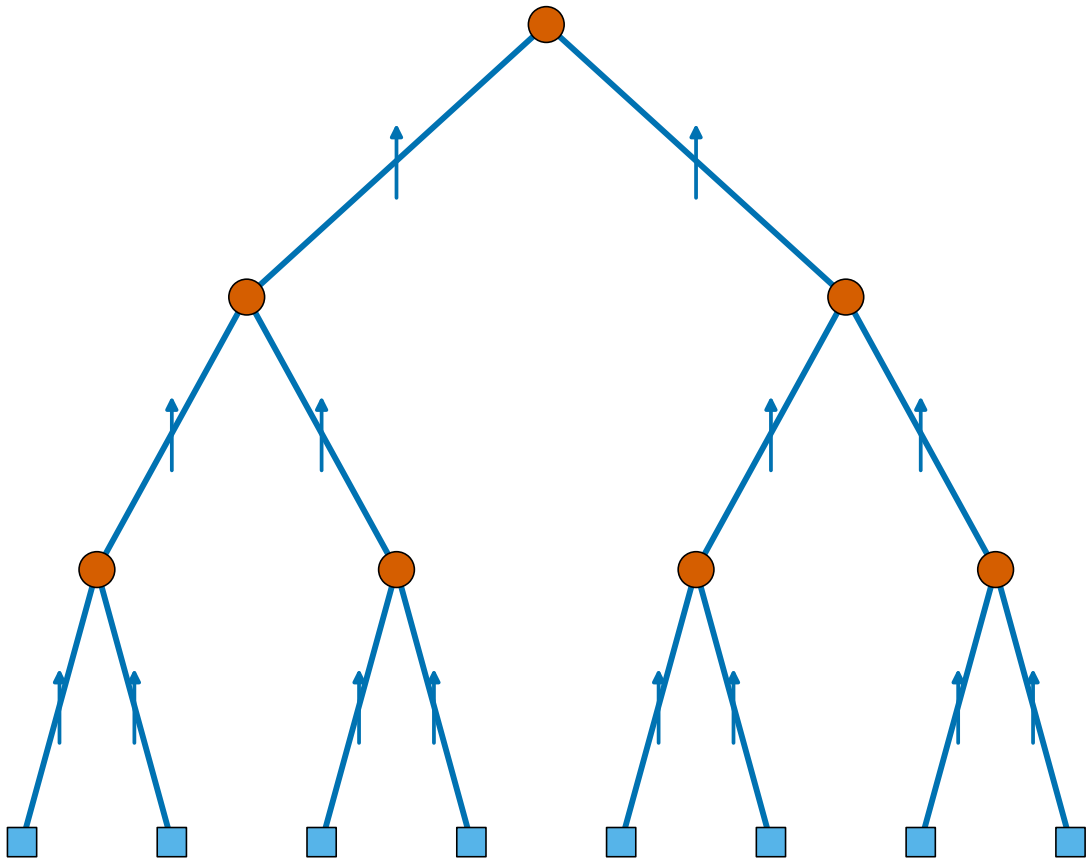
Merge, Bcast, Gate and LeafOut are the SAME tensors at every level and every node, so the model is defined on 'a separator tree', not on 'this grid'. Re-binding to a new system recompiles the schedule and reuses every weight -- which is what makes the zero-shot 14 -> 118 transfer of Part 9 possible at all.

1. bottom-up sweep

$$h_p = \text{LN Merge}[h_{lo}; h_{hi}; \bar{h}_{s_p}]$$

h_{root}

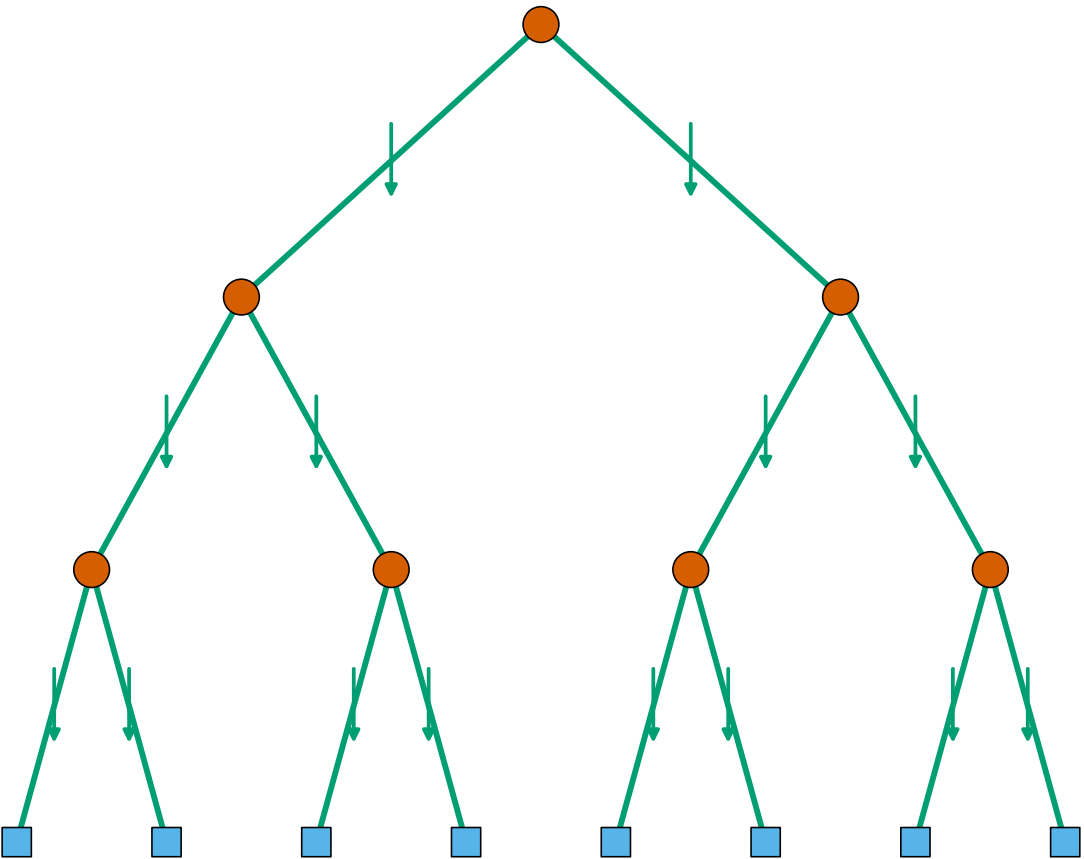
leaves = buses



information flows UP: the root ends up summarising the entire grid, through separators that are the physics bottlenecks

2. top-down sweep (x K)

$$h_c \leftarrow \text{LN}(h_c + g \odot \text{Bcast}[h_p; h_c])$$



information flows DOWN: each node re-reads its own state in the context of the whole grid. Repeating this is a truncated fixed-point (deep-equilibrium) iteration.